

Junghun Hwang (황정헌)

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Education

Konkuk University

Mar 2022 – Expected Feb 2027

B.S. in Smart Mobility Engineering | Seoul, Korea

Research & Project Experience

2026 9th Kookmin University Autonomous Driving Competition — AI-Edge Challenge

Apr 2026 – Aug 2026

Team Lead | Team of 5

- Built a ROS 2 imitation-learning pipeline covering driving-data collection, model training, TorchScript export, and real-time inference for XYCAR.
- Implemented hybrid driving logic that switched between learning-based control and deterministic rule-based traffic-light and mission handling.
- Managed the complete Roboflow dataset workflow, including upload, annotation, review, class organization, augmentation, split, and export, and directly trained a unified YOLO object-detection model for traffic lights, vehicles, obstacles, and cones.
- Implemented and deployed Mission Manager / Final Driver integration on the competition vehicle and contributed to real-vehicle system integration and parameter tuning.

Results: 2nd Place, Preliminary Round | 13th Place, Finals | Special Award — Parking Event

2026 Student AI · SW Mobility Competition

Mar 2026 – Ongoing

Team of 27 | Decision-Making Subteam (5)

- Co-developed a ROS 2 AEB driving pipeline, including cone-boundary path generation, red-entry stop logic, and stateful fail-safe behavior.
- Implemented Stanley-based path tracking with steering constraints and motion-preview compensation for cone-corridor driving.
- Validated the AEB pipeline in Gazebo closed-loop simulation and used Foxglove for path, steering, and stop-state visualization and debugging.
- Conducted initial real-vehicle testing at approximately 7 km/h and investigated steering oscillation observed during track tests.

Camera-Based V2I Cooperative End-to-End Autonomous Driving

2026 – Ongoing

Capstone Design Project

Phase 1: System & Experiment Design

Phase 2: Implementation & Closed-loop Validation (Planned)

- Compared smartphone, radar, LiDAR, and camera options for V2X infrastructure sensing, selecting a camera-based V2I architecture based on deployment feasibility and semantic information.
- Designed a blind-spot scenario with a pedestrian emerging between parked vehicles and defined a comparative evaluation between camera-only E2E driving and V2I-assisted E2E driving.
- Defined distance-to-hazard (d) as the primary experimental variable and designed quantitative evaluation using collision rate, minimum TTC, braking onset, stopping distance, and success rate.

ROS 2 UAV Waypoint Mission on PX4 SITL/Gazebo

May 2026 – Jun 2026

Autonomous System Platform Course Project

UAV Waypoint Development | Team of 4

- Developed a ROS 2 UAV waypoint-mission package that loaded 3D waypoints from CSV and advanced goals using TF-based pose tracking and configurable arrival tolerances.
- Integrated the mission node with an existing PX4 offboard controller through PoseStamped commands, supporting map/ENU and local-NED coordinate modes with mission lifecycle and safety logic.
- Successfully demonstrated full sequential waypoint tracking and final autonomous landing in the course project demonstration.

V-Model Neuro-Symbolic Driving

Mar 2026 – Jun 2026

Dream Semester Project | Decision-Making | Team of 3

- Implemented ROS 2 decision-making, waypoint-following, Pure Pursuit, and longitudinal speed-control components and integrated them with CARLA.
- Conducted closed-loop driving experiments and iteratively adjusted steering direction, lookahead, target smoothing, and command limits to address unstable path tracking.
- Although stable autonomous driving was not achieved, the project established foundational experience in ROS 2, CARLA, path tracking, and control integration that informed later autonomous-driving projects.

Awards & Competitions

2026 9th Kookmin University Autonomous Driving Competition — AI-Edge Challenge

2nd Place, Preliminary Round | 13th Place, Finals | Special Award — Parking Event

2022 Drone Clash, General Division

3rd Place

Technical Skills

Languages: Python, C++

Autonomous Systems: ROS 2, CARLA, Gazebo, PX4 SITL

AI / Computer Vision: PyTorch, YOLO, OpenCV, Roboflow

Tools / Environment: Ubuntu, Git/GitHub, Foxglove, RViz2, rosbag2, QGroundControl